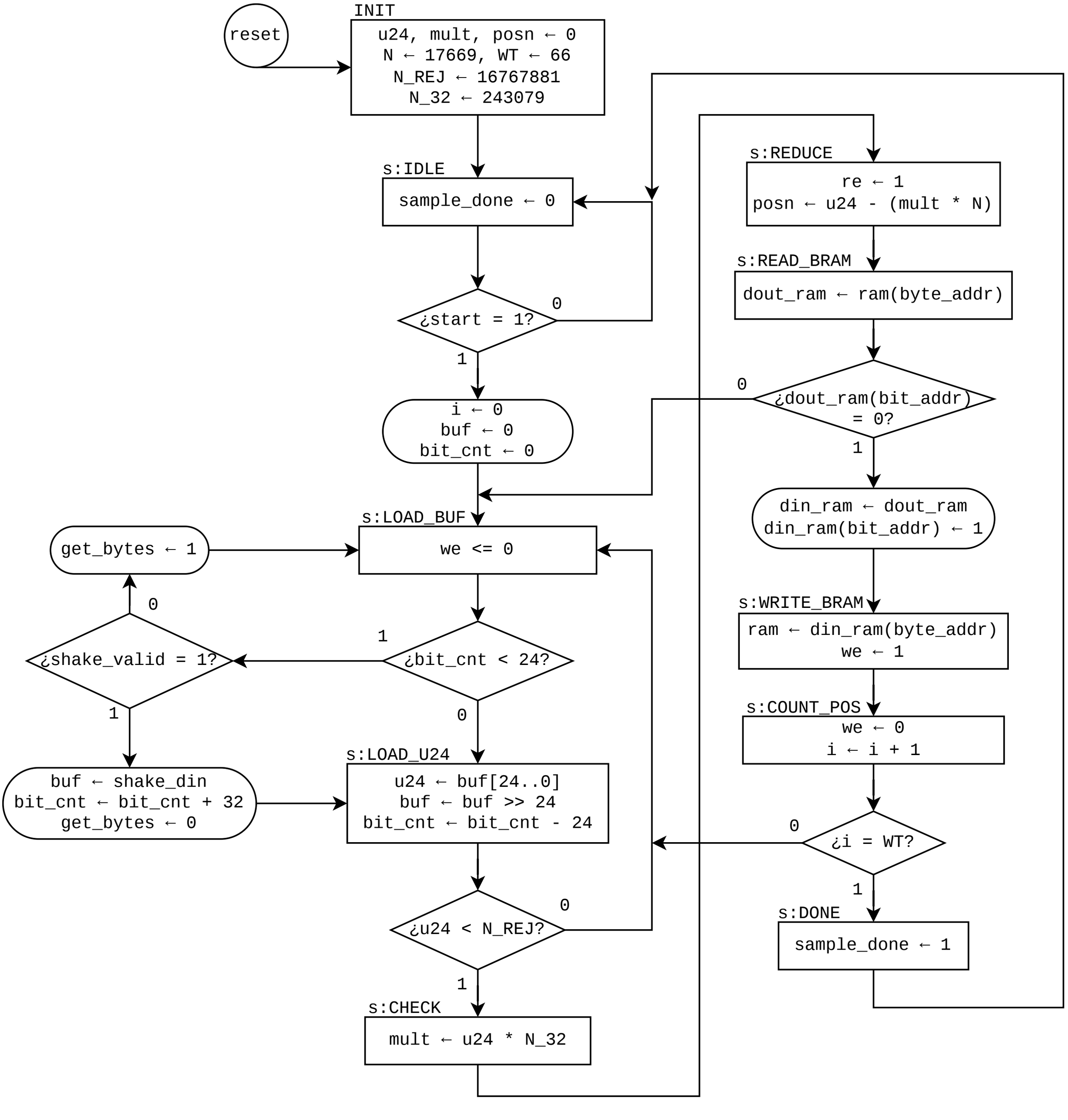
reset
INIT
u24, mult, posn ← 0
N ← 17669, WT ← 66
N_REJ ← 16767881
N_32 ← 243079
s:IDLE
sample_done ← 0
¿start = 1?
0
1
i ← 0
buf ← 0
bit_cnt ← 0
s:LOAD_BUF
get_bytes ← 1
we <= 0
0
¿shake_valid = 1?
1
¿bit_cnt < 24?
1
0
buf ← shake_din
bit_cnt ← bit_cnt + 32
get_bytes ← 0
s:LOAD_U24
u24 ← buf[24..0]
buf ← buf >> 24
bit_cnt ← bit_cnt - 24
¿u24 < N_REJ?
0
1
s:CHECK
mult ← u24 * N_32
s:REDUCE
re ← 1
posn ← u24 - (mult * N)
s:READ_BRAM
dout_ram ← ram(byte_addr)
¿dout_ram(bit_addr)
= 0?
0
1
din_ram ← dout_ram
din_ram(bit_addr) ← 1
s:WRITE_BRAM
ram ← din_ram(byte_addr)
we ← 1
s:COUNT_POS
we ← 0
i ← i + 1
¿i = WT?
0
1
s:DONE
sample_done ← 1